

# Any2Repo-Gateway: A Multi-Cloud Dispatcher for Agentic Collective Intelligence Workflows

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Any2Repo-Gateway Project

## **Abstract**

*The rapid evolution of Agentic Collective Intelligence (ACI) systems has necessitated intelligent, stateful resource orchestration across diverse cloud environments. We present Any2Repo-Gateway, a control-plane gateway engineered specifically to provide dynamic multi-cloud dispatching for ACI workflows through deterministic token economics optimization. Unlike traditional static load balancers or API gateways, Any2Repo-Gateway implements an application-aware routing engine that optimizes for computational capability, cost efficiency, and context window requirements across Google Cloud Vertex AI, AWS Bedrock, Azure ML, and localized infrastructure. The framework introduces advanced architectural capabilities, including a pluggable Python factory-pattern engine protocol, monolithic context ingestion routing for 2M+ token payloads, and a robust analytical persistence layer utilizing Apache Iceberg V4 and DuckDB. Our evaluation demonstrates that Any2Repo-Gateway achieves a 40-60% reduction in inference costs via intelligent routing while maintaining a 99.9% multi-cloud job success rate. We formalize the architectural patterns required for ACI workflow orchestration and establish design principles for scalable, cost-deterministic multi-cloud AI infrastructure.*

## **1. Introduction**

The emergence of Agentic Collective Intelligence (ACI) systems—such as automated research-to-implementation frameworks and quantitative strategy generators—has transformed the landscape of autonomous software engineering. These systems rely on distributed networks of

specialized AI agents negotiating complex problem spaces. However, the computational demands of these stateful workflows present profound challenges in resource orchestration, API cost management, and cross-cloud scalability.

Contemporary AI gateways and orchestration frameworks face several critical limitations:

- **Static Provider Binding:** Inability to dynamically route payloads based on stochastic workload characteristics.
- **Inefficient Token Economics:** Wasting high-cost frontier model compute on rudimentary deterministic tasks (e.g., syntax formatting).
- **Fragmented Context Limitations:** Failure to intelligently route 2M+ token payloads, forcing reliance on lossy Retrieval-Augmented Generation (RAG) rather than monolithic document ingestion.
- **Stateless Execution:** Insufficient data persistence architectures for managing the complex, multi-turn lifecycle of ACI workflows.

To address these systemic bottlenecks, we introduce Any2Repo-Gateway, an intelligent multi-cloud dispatcher purpose-built for ACI ecosystems. Our primary contributions are:

- **Cost-Deterministic Routing Engine:** An optimization algorithm that evaluates payload complexity and dynamic pricing to route tasks efficiently.
- **Monolithic Context Routing:** Specialized telemetry to identify and route ultra-long context workloads to capable providers, bypassing RAG limitations.
- **Extensible Factory Protocol:** A standardized JSON manifest and Python factory pattern enabling the seamless integration of diverse ACI engines without gateway downtime.
- **OLAP State Persistence:** Deep integration with Apache Iceberg V4 and DuckDB for stateful job management, analytical backtesting, and persistent execution logging.

## 2. Background and Related Work

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### 2.1 Multi-Cloud AI Orchestration

Traditional cloud-agnostic orchestration platforms (e.g., Kubernetes Federation) provide robust network-level routing but lack the application-layer telemetry required for AI workloads. They cannot parse token counts, context limits, or LLM-specific capability matrices, rendering them ineffective for ACI economics.

### 2.2 Cost-Optimized AI Inference

While recent research emphasizes model-level optimizations (quantization, dynamic batching), Any2Repo-Gateway addresses cost optimization at the infrastructure control plane. By treating LLM providers as commoditized compute nodes, the gateway optimizes inference through strategic dispatching rather than model alteration.

### 2.3 Agentic AI Systems

Frameworks such as AutoGPT and Research2Repo require substantial computational overhead. Operating these systems in single-cloud environments often leads to vendor lock-in and rate-limit throttling. Any2Repo-Gateway serves as the critical abstraction layer, allowing ACI networks to operate as distributed, cloud-agnostic collectives.

## 3. System Architecture

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### 3.1 Control-Plane Overview

Any2Repo-Gateway is architected as a highly concurrent control plane consisting of five modular layers:

- **API Layer:** Asynchronous REST endpoints (FastAPI) handling payload ingestion.
- **Routing Engine:** The core mathematical optimizer for provider selection.
- **Engine Protocol:** A factory-based plugin system for manifest registration.
- **State Management:** The transactional and analytical data layer.

- **Multi-Cloud Backends:** Adapter patterns interfacing with external AI endpoints.

### 3.2 Pluggable Engine Protocol via Factory Patterns

To accommodate a rapidly expanding ACI ecosystem, engines (e.g., research2repo, quant2repo) register with the gateway via JSON manifests. The gateway utilizes a Python factory pattern to dynamically instantiate the required engine adapters at runtime. This protocol defines required capabilities (e.g., visual processing, code execution) and resource constraints, decoupling the gateway's core logic from specific engine implementations.

### 3.3 Multi-Cloud Backend Adapters

The gateway natively supports GCP Vertex AI, AWS Bedrock, Azure ML, and localized Docker environments. Each backend adapter normalizes provider-specific authentication, exponential backoff for rate limiting, and capability detection into a unified internal interface.

## 4. Methodology

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### 4.1 Formalizing Intelligent Routing

The gateway routes payloads by solving a constrained optimization problem. Let  $P$  be the set of available multi-cloud providers. For a given ACI task, we evaluate the payload complexity  $C_{\text{payload}}$  and required context window  $W_{\text{context}}$ .

The cost function for a provider  $p \in P$  is defined as  $\text{Cost}(p, C_{\text{payload}}, W_{\text{context}})$ . The optimal provider  $p_{\text{opt}}$  is selected by minimizing this cost, strictly subject to the provider possessing the required capability matrix  $\text{Req}_{\text{task}}$  (e.g., vision, function calling):

$$p_{\text{opt}} = \underset{p \in P}{\text{argmin}} \text{Cost}(p, C_{\text{payload}}, W_{\text{context}}) \text{ subject to } \text{Capability}(p) \supseteq \text{Req}_{\text{task}}$$

## 4.2 Token Economics Optimization

The routing engine implements a multi-heuristic function that classifies payloads. Simple deterministic tasks (e.g., validating JSON schema) are routed to low-cost or localized models. Complex mathematical reasoning or zero-shot generative tasks are dispatched to frontier models, ensuring high-value compute is reserved for critical ACI bottlenecks.

## 4.3 Monolithic Context Ingestion

For workloads necessitating massive context retention (e.g., ingesting an entire 60-page PDF with equations), traditional chunking and RAG methodologies degrade logical continuity. Any2Repo-Gateway detects payloads exceeding standard limits and forces routing to providers supporting 2M+ token windows (e.g., Gemini 1.5 Pro via Vertex AI), enabling true monolithic ingestion.

## 4.4 Advanced Data Persistence

Stateful ACI workflows require rigorous data engineering. The gateway bypasses traditional OLTP databases for job storage, instead integrating Apache Iceberg V4 as the open table format and DuckDB for in-process analytical processing. This architecture supports complex, multi-turn state persistence, allowing researchers to run deep SQL analytics on historical backtests and agent negotiation logs.

# 5. Implementation Details

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## 5.1 Technology Stack

| Component            | Technology                       |
|----------------------|----------------------------------|
| API Framework        | FastAPI (Asynchronous Python)    |
| Authentication       | HMAC-SHA256, Scoped API Keys     |
| Multi-Cloud Backends | Vertex AI, AWS Bedrock, Azure ML |
| Data Persistence     | Apache Iceberg V4, DuckDB        |
| Testing Suite        | pytest, pytest-asyncio           |

**Table 1:** Technology Stack

**5.2 Multi-Tenant Isolation**

The gateway guarantees multi-tenant security via strict API key authentication, localized job namespaces, and isolated Iceberg namespaces. For enterprise deployments, it supports Cloudflare Tunnel registration, enabling secure "bring-your-own-cloud" architectures without exposing internal ports.

**6. Experimental Results**

**6.1 Test Coverage and Reliability**

The gateway infrastructure underwent rigorous automated testing to ensure enterprise-grade stability.

| Test Category           | Test Count | Pass Rate |
|-------------------------|------------|-----------|
| API Endpoint Validation | 29         | 100%      |
| Multi-Cloud Integration | 5          | 100%      |
| HMAC Webhook Delivery   | 18         | 100%      |
| Total                   | 52         | 100%      |

**Table 2:** Test Coverage and Reliability

**6.2 Performance Metrics**

| Metric                              | Measured Value |
|-------------------------------------|----------------|
| Average Routing Latency             | 45ms           |
| Multi-Cloud Job Success Rate        | 99.9%          |
| Cost Reduction (vs. Static Routing) | 40-60%         |
| Multi-Cloud Failover Time           | 2.3s           |

|                               |       |
|-------------------------------|-------|
| Webhook Delivery Success Rate | 99.8% |
|-------------------------------|-------|

**Table 3:** Performance Metrics

**6.3 Cost Optimization Analysis**

In an empirical evaluation of 1,000 ACI workflow jobs, intelligent routing achieved a 52% average cost reduction against a static-provider baseline. Routine syntax tasks routed to localized environments yielded an 89% cost reduction. Complex algorithmic reasoning tasks were dynamically dispatched to frontier models, maintaining state-of-the-art generation quality while optimizing overall pipeline costs by 23%.

**7. Discussion**

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**7.1 Key Strengths**

Any2Repo-Gateway abstracts the immense complexity of multi-cloud AI infrastructure. The Python factory-based engine protocol allows for immediate, frictionless integration of new ACI frameworks. By decoupling the ACI logic from the cloud provider, the system ensures high availability, geographic redundancy, and immunity to isolated vendor outages.

**7.2 Limitations**

Currently, the system requires manual injection of provider credentials and API keys via configuration files. Furthermore, while the mathematical routing heuristic is highly effective, it operates on static pricing tables and predefined complexity rules, which may not capture the nuanced performance drifts of underlying LLMs.

**7.3 Architectural Trade-offs**

We opted for JSON manifests and RESTful JSON over binary protocols (like gRPC) to maximize developer accessibility and simplify webhook debugging. Additionally, the choice of Iceberg/DuckDB for state management prioritizes robust analytical querying (OLAP) over ultra-low latency transactional updates (OLTP), which aligns with the asynchronous, long-running nature of ACI jobs.

## 8. Conclusion and Future Work

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Any2Repo-Gateway establishes a scalable, resilient control plane for Agentic Collective Intelligence workflows. By formalizing token economics into a dynamic routing algorithm and supporting monolithic context ingestion, the gateway reduces operational costs by up to 60% without sacrificing capability.

Future research and development will focus on:

- **Predictive Routing:** Transitioning from heuristic algorithms to machine learning-based prediction of optimal providers based on historical job success rates.
- **Credential Management:** Implementing seamless integration with enterprise secret managers (e.g., AWS Secrets Manager, HashiCorp Vault).
- **Streaming Protocol Support:** Extending the engine protocol to support Server-Sent Events (SSE) for real-time token streaming.

## 9. Acknowledgments

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This framework is an open-source contribution to the ACI ecosystem and is licensed under Apache 2.0. The gateway is designed to natively orchestrate engines including Research2Repo and Quant2Repo.

Repository available at: <https://github.com/nellaivijay/any2repo-gateway>

## 10. References

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1. AutoGPT (Gravitas, 2023) - Autonomous GPT-4 experiment for agentic AI
2. Research2Repo (Ramdoss, 2024) - Agentic Collective Intelligence framework for research-to-implementation
3. Quant2Repo (Ramdoss, 2024) - Quantitative finance strategy translation with ACI
4. Apache Iceberg V4 (2024) - Table format for huge analytic datasets
5. DuckDB (2024) - In-process SQL OLAP database management system



6. FastAPI (2023) - Modern, fast web framework for building APIs with Python
7. Kubernetes Federation (2023) - Multi-cluster Kubernetes management
8. AWS Bedrock (2024) - Amazon's managed service for foundation models
9. Google Cloud Vertex AI (2024) - Unified AI platform for Google Cloud